

# Jaihoon Kim

[Website](#) | [LinkedIn](#) | Email: jh27kim@kaist.ac.kr

## RESEARCH INTERESTS

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Generative AI, Diffusion Model, Flow Model, Discrete Diffusion

## EDUCATION

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|                                                            |                                |
|------------------------------------------------------------|--------------------------------|
| Korea Advanced Institute of Science and Technology (KAIST) | Sep 2024 – Feb 2028 (Expected) |
| Ph.D. in Computer Science                                  | Advisor: Minhyuk Sung          |
| Daejeon, South Korea                                       |                                |

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|------------------------------------------------------------|-----------------------|
| Korea Advanced Institute of Science and Technology (KAIST) | 2023 – Aug 2024       |
| M.S. in Computer Science                                   | Advisor: Minhyuk Sung |
| Daejeon, South Korea                                       |                       |

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|--------------------------------|-------------|
| Hanyang University             | 2015 – 2021 |
| B.S. in Electrical Engineering |             |
| Seoul, South Korea             |             |

## WORK EXPERIENCE

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|                                    |                      |
|------------------------------------|----------------------|
| Netflix (Los Gatos, United States) | June 2026 – Sep 2026 |
| Research Scientist (Internship)    |                      |

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| Amazon Web Services (Santa Clara, United States) | July 2025 – Nov 2025 |
| Research Scientist (Internship)                  |                      |

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|--------------------------------------------|----------------------|
| Samsung Electronics (Hwasung, South Korea) | July 2021 – Jan 2023 |
| Cloud Engineer (Full Time)                 |                      |

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|--------------------------------------------|-----------------------|
| Samsung Electronics (Hwasung, South Korea) | July 2020 – July 2020 |
| Cloud Engineer (Internship)                |                       |

## PUBLICATIONS

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1. NoiseTilt: Noise-Tilted Reverse Kernels for Diffusion Reward Alignment  
Jisung Hwang, Yunhong Min, **Jaihoon Kim**, I-Chao Shen, Minhyuk Sung  
ECCV 2026
2. MatLat: Material Latent Space for PBR Texture Generation  
Kyeongmin Yeo, Yunhong Min, **Jaihoon Kim**, Minhyuk Sung  
CVPR 2026 (Highlight)
3. Demystifying Transition Matching: When and Why It Can Beat Flow Matching  
**Jaihoon Kim**, Rajarshi Saha, Minhyuk Sung, Youngsuk Park  
AISTATS 2026
4. Inference-time scaling for flow models via stochastic generation and rollover budget forcing  
**Jaihoon Kim**\*, Taehoon Yoon\*, Jisung Hwang\*, Minhyuk Sung (\* equal contribution)  
NeurIPS 2025
5. Moment-and Power-Spectrum-Based Gaussianity Regularization for Text-to-Image Models  
Jisung Hwang, **Jaihoon Kim**, Minhyuk Sung  
NeurIPS 2025

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6. Moment-and Power-Spectrum-Based Gaussianity Regularization for Text-to-Image Models  
Prin Phunyaphibarn, Phillip Y Lee, **Jaihoon Kim**, Minhyuk Sung  
WACV 2026
7. StochSync: Stochastic Diffusion Synchronization for Image Generation in Arbitrary Spaces  
Kyeongmin Yeo\*, **Jaihoon Kim\***, Minhyuk Sung (\* equal contribution)  
ICLR 2025
8. SyncTweedies: A General Generative Framework Based on Synchronized Diffusions  
**Jaihoon Kim\***, Juil Koo\*, Kyeongmin Yeo\*, Minhyuk Sung (\* equal contribution)  
NeurIPS 2024
9. Paired-Point Lifting for Enhanced Privacy-Preserving Visual Localization  
Chunghwan Lee, **Jaihoon Kim**, Chanhyuk Yun, Je Hyeong Hong  
CVPR 2023

## PREPRINTS

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1. Contrastive Distribution Matching for Amortized Sequential Monte Carlo in Discrete Diffusion  
**Jaihoon Kim**, Taehoon Yoon, Prin Phunyaphibarn, Seungjun Kim, Morteza Mardani, Minhyuk Sung  
arXiv 2026

## TEACHING EXPERIENCE

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|                                                                       |                                           |
|-----------------------------------------------------------------------|-------------------------------------------|
| CS479: Machine Learning for 3D Data<br>Course Assistant               | School of Computing, KAIST<br>Spring 2025 |
| CS492(D): Diffusion Models and Their Applications<br>Course Assistant | School of Computing, KAIST<br>Fall 2024   |
| CS580: Computer Graphics<br>Course Assistant                          | School of Computing, KAIST<br>Spring 2024 |
| CS479: Machine Learning for 3D Data<br>Course Assistant               | School of Computing, KAIST<br>Fall 2023   |

## SCHOLARSHIPS AND ACHIEVEMENTS

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|                                                                                       |             |
|---------------------------------------------------------------------------------------|-------------|
| Master's Degree Outstanding Paper<br>KAIST                                            | Summer 2024 |
| Grand Prix (2 <sup>nd</sup> place) x, X-Corps Festival<br>Ministry of Science and ICT | Spring 2020 |
| Honors<br>University of Texas at Austin                                               | Fall 2019   |
| Dean's List<br>Hanyang University                                                     | Spring 2019 |

## ACADEMIC SERVICE

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Reviewer  
ICLR 2024, 2025, 2026  
NeurIPS 2024, 2025, 2026

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ICCV 2025, ECCV 2026  
CVPR 2026

Pacific Graphics 2023  
Student Delegate

Fall 2023

## SKILLS

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|                       |                                   |
|-----------------------|-----------------------------------|
| Programming Languages | Python, C++                       |
| Frameworks            | PyTorch                           |
| Languages             | Korean (Native), English (Fluent) |